

AI/ML PROJECT SUBMISSION



PROJECT OVERVIEW	STUDENT DETAILS
• Project Number: 4 • Task Name: Quiz and Exam System • Difficulty: Moderate	• Name: Pranav Gajanan Rane • College: Government Polytechnic Gondia • Branch: Computer Engineering • Domain: AI/ML

Project 3 : Quiz & Exam System

The Quiz and Examination System is an interactive Python application that presents multiple choice questions (MCQs) to a student, evaluates responses, tracks scores, and generates a detailed result report at the end. The question bank is stored using tuples (immutable, for data integrity) and the application handles timing, scoring, and grade calculation automatically.

CODE-

```
#-----  
-----  
'''
```

Project 4 – Quiz & Examination System

Made by Pranav Rane

```
'''
```

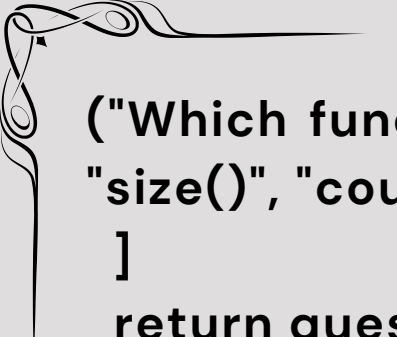
```
#-----  
-----
```

'''Functions Definitions followed by actual system.'''

import random # Used to shuffle our questions each time

def load_questions(): # Questions for the quiz is stored here with the Answers

```
    questions = [  
        ("Which data type is immutable in Python?", "List",  
"Dictionary", "Tuple", "Set", "C"),  
        ("What is the output of print(2 ** 3)?", "6", "8", "9",  
"12", "B"),  
        ("Which keyword is used to define a function in  
Python?", "func", "define", "def", "function", "C"),  
        ("What does len('hello') return?", "4", "5", "6", "error",  
"B"),  
        ("Which of the following is a mutable data type?",  
"Tuple", "String", "List", "Integer", "C"),  
        ("What symbol is used for single-line comments in  
Python?", "//", "/* */", "#", "--", "C"),  
        ("What will type(3.14) return?", "int", "float", "str",  
"double", "B"),  
        ("How do you take input from the user in Python?",  
"scan()", "cin()", "input()", "readline()", "C"),  
        ("What does the append() method do to a list?",  
"Removes last element", "Adds element at beginning",  
"Adds element at end", "Sorts the list", "C"),  
        ("Which loop is best when number of iterations is  
already known?", "while", "do-while", "for", "repeat",  
"C"),  
        ("What is the index of the first element in a Python  
list?", "-1", "1", "0", "None", "C"),
```



```
("Which function is used to find the length of a list?",  
"size()", "count()", "len()", "length()", "C"),  
]  
return questions
```

```
def display_question(qno, q): # Used to display the  
question with their options  
    print(f"Q{qno}: {q[0]}")  
    print("OPTIONS - ")  
    print(f" A) {q[1]}")  
    print(f" B) {q[2]}")  
    print(f" C) {q[3]}")  
    print(f" D) {q[4]}")
```

```
def get_answer(): # To take answer as an input from  
the user  
    while True:  
        try:  
            ans = input("Your Answer: ").upper()  
            if ans in ["A", "B", "C", "D"]:  
                return ans  
            else:  
                print("Please enter A, B, C or D options only.")  
        except Exception:  
            print("Invalid input. Try again.")
```

```
def calculate_grade(percentage): # Showcases the  
percentage
```



```
if percentage >= 90:
    return "O"
elif percentage >= 80:
    return "A+"
elif percentage >= 70:
    return "A"
elif percentage >= 60:
    return "B+"
elif percentage >= 50:
    return "B"
else:
    return "F"
```

```
def show_wrong_answers(wrong_list): # Displays
Wrong answers choosen
    if len(wrong_list) == 0:
        print("\nNo wrong answers. Great job!")
        return
```

```
print("\n--- Wrong Answers Review ---")
for entry in wrong_list:
    qno = entry[0]
    q = entry[1]
    your_ans = entry[2]
    options = {"A": q[1], "B": q[2], "C": q[3], "D": q[4]}
    correct = q[5]
    print(f"\Q{qno}: {q[0]}")
        print(f"    Your Answer : {your_ans} -
{options[your_ans]}")
        print(f"    Correct Ans : {correct} -
{options[correct]}")
```

```
def show_report(name, roll, score, total, wrong_list):
# Shows the final report for the quiz
    percentage = (score / total) * 100
    grade = calculate_grade(percentage)

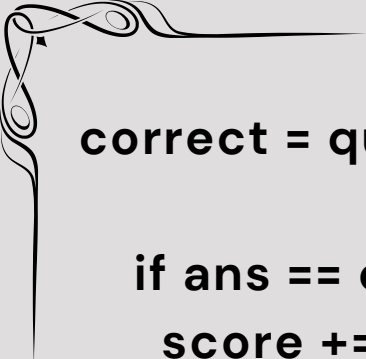
    if percentage >= 50:
        result = "PASS"
    else:
        result = "FAIL"

    print("\n===== QUIZ RESULT =====")
    print(f"Name : {name}")
    print(f"Roll No : {roll}")
    print(f"Score : {score} / {total}")
    print(f"Percentage : {percentage:.2f}%")
    print(f"Grade : {grade}")
    print(f"Result : {result}")
    print("=====")

    show_wrong_answers(wrong_list)
```

```
def evaluate_quiz(name, roll, questions): # Initializes
the quiz
    score = 0
    wrong_list = []
    total = len(questions)

    for i in range(total):
        display_question(i + 1, questions[i])
        ans = get_answer()
```



```
correct = questions[i][5]
```

```
if ans == correct:
```

```
    score += 1
```

```
    print("Correct!")
```

```
else:
```

```
    print(f"Wrong! Correct answer was: {correct}")
```

```
    wrong_list.append((i + 1, questions[i], ans))
```

```
show_report(name, roll, score, total, wrong_list)
```

```
def main(): # main method
```

```
    print("====++++PYTHON QUIZ++++====")
```

```
    name = input("Enter your name: ")
```

```
    roll = None
```

```
    while roll is None: # Essential for one time use
```

```
        try:
```

```
            roll = int(input("Enter Roll Number: "))
```

```
        except ValueError:
```

```
            print("Roll number should be an integer.")
```

```
    print(f"Student: {name} | Roll: {roll}")
```

```
    questions = load_questions()
```

```
    random.shuffle(questions)
```



```
print(f"Total Questions Available: {len(questions)}")
print("Enter choice A, B, C or D for each question.")

evaluate_quiz(name, roll, questions)
```

```
main()
```

Note - This code might break due to incorrect indentation while pdf making so pls use the code mentioned in .py file attached

TEST OUTPUTS -

```
====++++PYTHON QUIZ++++====
Enter your name: Pranav
Enter Roll Number: 15
Student: Pranav | Roll: 15
Total Questions Available: 12
Enter choice A, B, C or D for each question.
Q1: What will type(3.14) return?
OPTIONS -
  A) int
  B) float
  C) str
  D) double
Your Answer: b
Correct!
```

```
Q2: What symbol is used for single-line comments in Python?
OPTIONS -
  A) //
  B) /* */
  C) #
  D) --
Your Answer: c
Correct!
```


Q11: Which loop is best when number of iterations is already known?

OPTIONS -

- A) while
- B) do-while
- C) for
- D) repeat

Your Answer: d

Wrong! Correct answer was: C

===== QUIZ RESULT =====

Name : Pranav

Roll No : 15

Score : 9 / 12

Percentage : 75.00%

Grade : A

Result : PASS

=====

--- Wrong Answers Review ---

\Q7: Which of the following is a mutable data type?

Your Answer : A - Tuple

Correct Ans : C - List

\Q11: Which loop is best when number of iterations is already known?

Your Answer : D - repeat

Correct Ans : C - for

\Q12: How do you take input from the user in Python?

Your Answer : A - scan()

Correct Ans : C - input()

TEST OUTPUTS IN BROKEN CASES -

```
Q1: Which loop is best when number of iterations is already known?
OPTIONS -
A) while
B) do-while
C) for
D) repeat
Your Answer: 65
Please enter A, B, C or D options only.
Your Answer: 
```

```
===== QUIZ RESULT =====
Name       : Yash
Roll No    : 98
Score      : 0 / 12
Percentage : 0.00%
Grade      : F
Result     : FAIL
=====
```

CODE TESTING WAS DONE SUCCESSFULLY!

More Details about the project is mentioned in the .md file attached . This pdf was made specifically to attach the code and its outputs.

PROJECT COMPLETED

